



LoRaWAN based internet of things (IoT) system for precision irrigation in plasticulture fresh-market tomato

Haozhe Zhang^a, Long He^{a,b,*}, Francesco Di Gioia^c, Daeun Choi^a, Antonio Elia^d, Paul Heinemann^a

^a Department of Agricultural and Biological Engineering, Pennsylvania State University, University Park, PA 16802, USA

^b Penn State Fruit Research and Extension Center, 290 University Drive, Biglerville, PA 17307, USA

^c Department of Plant Sciences, Pennsylvania State University, University Park, 16802, USA

^d Department of Agriculture, Food, Natural Resources and Engineering, University of Foggia, Foggia, Italy

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ABSTRACT

Precision irrigation with sensors has proven to be effective for water saving in crop production. An Internet of Things (IoT) system is necessary for monitoring real-time data from sensors and automating irrigation systems. Long-range wide-area network (LoRaWAN), a type of low-power wide-area network (LPWAN), is a low-cost and easily implemented IoT system which can be used for precision crop irrigation. In this study, a LoRaWAN based IoT system was developed and evaluated as a precision irrigation tool on fresh-market tomato production in a plasticulture system. Four irrigation scheduling treatments were designed and tested, including irrigation based on crop evapotranspiration (ET), soil matric potential sensors (Watermark 200SS-5) at -60 kPa (MP60) or -40 kPa (MP40), and GesCoN fertigation decision support system (DSS). The treatments were arranged based on a randomized complete block design with four replicates. System feasibility, irrigation water use efficiency (iWUE), and crop yield were evaluated. Throughout the season, the overall water use efficiencies were 22.2, 26.5, 27.9, and 28.4 kg·m⁻³ for ET, MP60, MP40, and GesCoN, respectively. The results indicated that treatment MP60 and GesCoN had 15.2 and 22.1% higher marketable fruit yield than ET, while, MP40 had 12.5% lower marketable yield compared to ET. Overall, the LoRaWAN-based IoT system performed well in terms of power consumption, communication, sensor reading, and valve control. These results suggested that the IoT system can be implemented for precision and automatic irrigation operations for vegetable and other horticultural crops with enhancing crop's water use efficiency and sustainability.

1. Introduction

In the United States, agriculture is a major consumer of ground and surface water, accounting for approximately 80% of the nation's consumptive water use, and this percentage can be higher in the western states characterized by a dryer climate [1]. As the global population continues to increase, food-crop production is expected to increase dramatically while water resources are increasingly limited [2,3]. It is important to use water efficiently for crops such as vegetables, characterized by shallow roots and relatively high water content, and thus very sensitive to water stress [4,5]. Water can be in excess or deficit, and thus irrigation water management can substantially impact vegetable yield and quality [6,7].

Conventionally, farm managers determine when and how much to irrigate primarily based on their experiences and time availability, which often led to inefficient water usage and reduction in crop yield and qual-

ity either by over-irrigating or under-irrigating. Precision irrigation is defined as a modern irrigation management strategy that allows growers to avoid plant water stress at critical growth stages by applying only the necessary amount of water directly to the crop, with varying rate and duration as needed based on site-specific conditions [8]. By applying precision irrigation on agricultural crops, farmers are expected to benefit from lower costs of irrigation water and manpower, while improving crop yield and quality. Adoption of precision irrigation for crop production systems requires the development of integrated sensing, decision-making strategies, and control systems, eventually to precisely control the timing, rate and distribution of water as needed [9].

Precision irrigation can be applied based on soil moisture, plant stress, and evapotranspiration (ET) [10]. Among these methods, soil moisture sensor-based precision irrigation has been widely tested and used in vegetable fields and protected culture systems. Soil volumetric water content (VWC) and soil matric potential (MP) are two indi-

* Corresponding author at: Department of Agricultural and Biological Engineering, Pennsylvania State University, University Park, PA 16802, USA.
E-mail address: luh378@psu.edu (L. He).

cators for available water in the soil which can be used to implement soil moisture-based irrigation systems [11,12]. VWC is the ratio of water volume and soil volume with a unit of percentage. MP describes the force with which soil matrix holds water [13]. MP measures the required energy for water movement in the soil which is always a negative number expressed in kPa. Based on our preliminary study, soil MP sensors were selected for this study for their higher compatibility with internet of things (IoT) systems and higher sensitivity to variations of soil moisture, which is very critical for vegetable crops irrigation management [14,15].

A wired or wireless sensor network is one of the key technologies for precision and automated irrigation systems [16]. IoT is a combination of networked sensors and machines for capturing, transmitting, managing, and analyzing data. The data from sensors are first uploaded wirelessly to a server. Data are then available on the internet for analysis and computing. Finally, the server sends commands wirelessly to the actuators in the field for executing tasks. Various wireless technologies in IoT systems have been investigated for crop irrigation management, such as Wi-Fi, cellular network (GPRS, LTE), ZigBee, and LoRaWAN. Zhao et al. [17] compared the performance of Wi-Fi, ZigBee, GPRS, and LoRaWAN for irrigation systems. The results indicated that Wi-Fi and ZigBee had low coverage and only worked for the vegetable fields near the gateway. GPRS is good for long-distance communication, but it has high-power consumption, and high cost of maintenance and deployment. LoRaWAN technology has a maximum range of 10 km with low power consumption and cost, which could allow small farms to have affordable precision irrigation. This technology was originally published in 2015 by LoRa Alliance, a non-profit association supporting the LoRaWAN protocol. Goap et al. [18] developed a Wi-Fi based IoT smart irrigation management system with a machine learning algorithm to improve irrigation control. Some studies have reported the LoRaWAN-based architecture and the potential of implementation in smart irrigation [19–21]. A project called SWAMP tested the effect of their LoRaWAN IoT-based irrigation system at four pilot locations in Brazil and Europe [22]. While LoRaWAN based IoT technology has not been widely tested in precision irrigation systems for vegetable crops with soil moisture monitoring, valve control, and water usage efficiency analysis.

Therefore, the primary goal of this study was to develop an effective IoT-based precision irrigation system using LoRaWAN technology for vegetable production. The specific objectives were to: (1) develop a LoRaWAN technology-based IoT wireless sensing network system for precision irrigation for vegetable irrigation, (2) conduct a performance evaluation of the irrigation system in terms of data communication, irrigation execution, and power consumption, and (3) evaluate the efficacy of the IoT-based precision irrigation system at field scale using fresh-market tomato as a test crop, and compare it to other irrigation approaches such as evapotranspiration based method and a decision support system.

2. Materials and methods

2.1. Experimental design

An open field irrigation management experiment was conducted at the Horticultural Research Farm of the Penn State Russell E. Larson Agricultural Research Center (Furnace, PA) during the 2020 growing season. The soil type in the experimental field was silty clay loam soil with 13.5% of sand, 47.8% of silt, and 38.8% of clay. Soil pH was 6.5, and the initial contents of phosphate, potash, magnesium, and calcium was equivalent to 175, 298, 584, and 3590 kg ha⁻¹, respectively. Total nitrogen and carbon soil contents were 0.13% and 1.13%, respectively. The content of ammonium nitrogen was 2.34 mg/kg.

Fresh-market tomato (*Solanum lycopersicum* L.) cv. Red Deuce F1 (HM Clause, Davis, CA) was used as the test crop. Seedlings were planted at the fourth true-leaf stage on May 21st, 2020, on raised beds mulched with black polyethylene film served by a single drip tape per bed. Beds



Fig. 1. The tomato open field and irrigation system setup for the irrigation experiment.

were 0.91 m wide and spaced at 2.13 m, and plants were planted at an in-row distance of 0.46 m, establishing a density of 1.03 plants per m². The 16 mm diameter drip tape had emitters spaced at 0.30 m and had a flow rate of 1.0 L per hour per emitter at 55 kPa (T-Tape, Rivulis Irrigation Ltd. San Diego, CA). Plants were trellised using the stake and Florida weave method [23], and the crop was managed according to local practices using an integrated pest management approach. A view of the open field experiment is shown in Fig. 1.

Four treatments using different irrigation scheduling were designed and tested. Treatment 1 (denoted ET) was based on crop evapotranspiration using a dual crop coefficient model according to FAO Paper 56 [24]. Treatments 2 and 3 were based on soil matric potential (MP) sensors (Watermark 200SS-5, Irrrometer Company, Inc., Riverside, CA) at different thresholds of -60 kPa (denoted MP60) and -40 kPa (denoted MP40), respectively. Once the average sensor readings at 20 cm depth reached the thresholds, the IoT system would send a notification and valves were manually controlled through the IoT system interface. Treatment 4 (denoted GesCoN) was based on recommendations provided by a decision support system (DSS) named GesCoN [25,26] using the web-platform EcoFert (<https://www.ecofert.it/>) and their smart phone application [27,28].

The layout of the experimental field is shown in Fig. 2. The overall field (planting area) was 27.74 × 42.67 m. The field consisted of twelve raised beds. Three long beds constitute a block, each block was divided into four sections, and each section hosted one irrigation strategy treatment (T1 to T4). In overall, the field was divided into 16 experimental sections of the same size consisting of three 9.14 m beds. A space of 1.52 m served as a buffer zone to separate different treatments and replications. Twenty plants were placed in each bed, for a total of 60 plants per experimental section. Treatments were arranged in a randomized complete block design (RCBD) with four replicates. Four treatments were randomly distributed in each block. The crop was managed according to local standards and following integrated pest management practices.

2.2. Irrigation system setup

Fig. 3 shows the structure of the irrigation system, including pipelines, valves, pressure sensors, flow meters, and other accessories. The water was supplied from the main water line of the farm. The main irrigation line after a screen filter was divided into four pipelines, one for each treatment. Four latching DC solenoid valves (PGV Series 2.54 cm, Hunter Inc., San Marcos, CA) were installed at the beginning of each line, followed by the pressure sensors (G1/4, 5 V, 0–1.2 MPa) installed at each line behind the solenoid valves to measure the water pressure. The pressure indicated the status of the valves. The pressure regulators (241 kPa) behind the pressure sensors limited the water pressure reaching the following components. The flow meters measured the cumulative flow volume during the experiment. The fertigation system

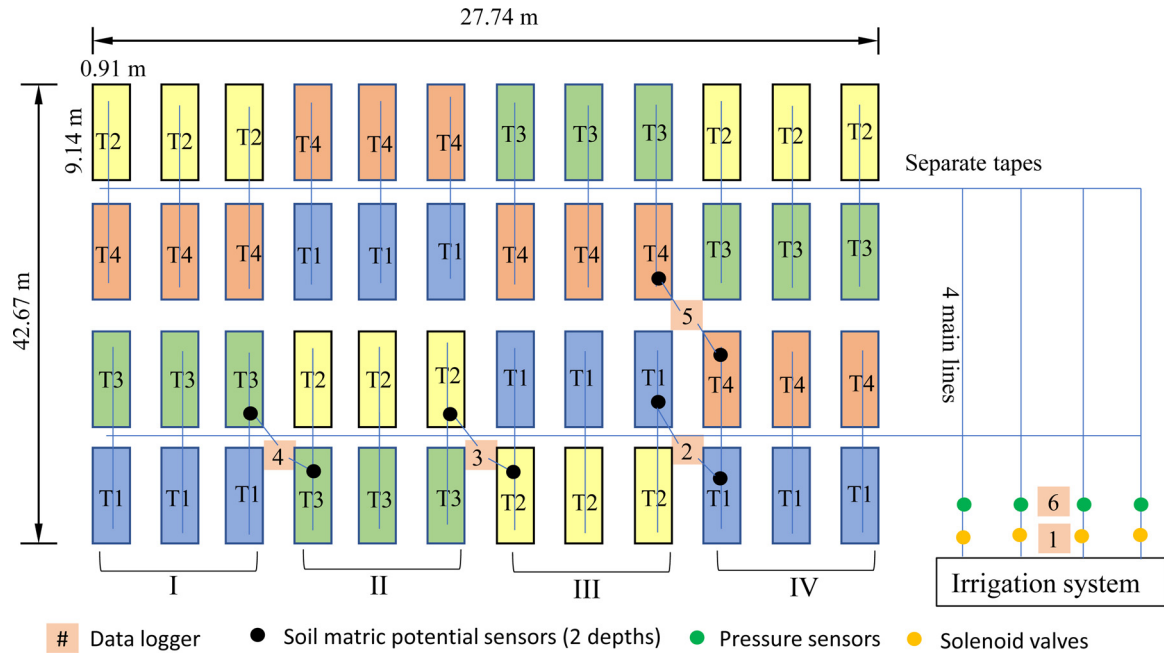


Fig. 2. Experimental setup and the locations of the sensors, valves, and data loggers. Plots marked as T1, T2, T3, and T4 were managed based on ET, MP60, MP40, and GesCoN treatment, respectively.

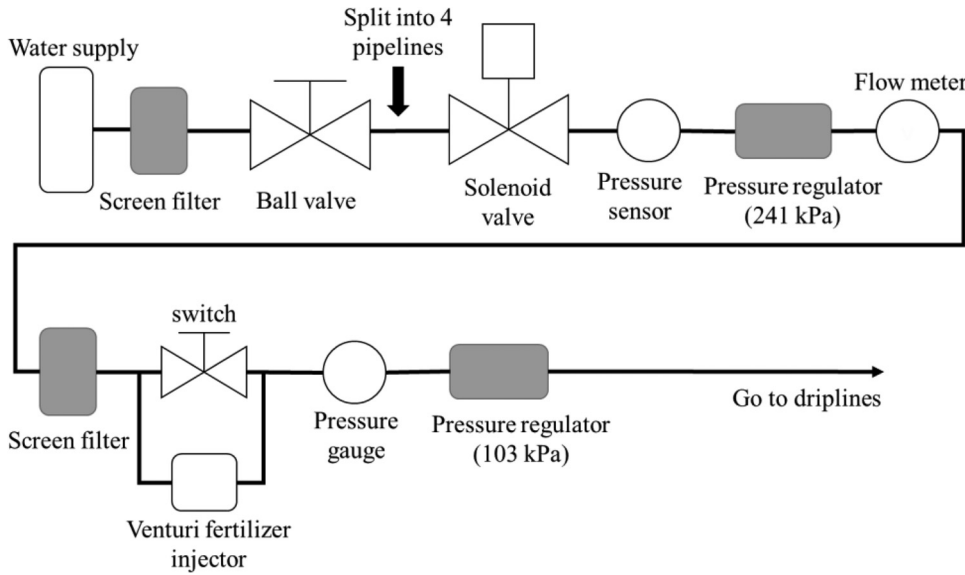


Fig. 3. A schematic diagram of developed irrigation system setup.

constituted by a by-pass with a Venturi injector was used to apply fertilizers when recommended by the GesCoN. When fertigation started, the injector was opened first, then the switch in the pipeline was closed. When fertigation ended, the switch in the pipeline was opened first, then the injector was closed. The pressure gauges were used to measure the real-time water pressure and calibrate pressure sensors. The last components were pressure regulators at 103 kPa, which was the suggested pressure for the irrigation drip tapes. Drip tapes were placed underneath the mulch for each section. Then the drip lines on the same treatment were connected to the pipes in the field pathway. Finally, these pathway pipes were connected to the corresponding pipelines (2.54 cm) for each treatment.

2.3. Sensor and valve systems setup

As indicated in Fig. 2, soil matrix potential (MP) sensors, pressure sensors, and valves were connected to six data loggers in the irrigation system. Fig. 4 illustrates the installation of soil moisture sensors, solenoid valves, pressure sensors, and data loggers in the field. The major components of the data loggers included a base control board (Vinduino LLC, Temecula, CA) and a LoRaWAN wireless Module (LM513-H, GlobalSat WorldCom Corp., New Taipei City, Taiwan) with an antenna. Each data logger was powered by a 3.7 V lithium-ion polymer (LiPo) battery, and a solar panel (70 × 70 mm) was connected to charge the battery. Data loggers 1 and 6 used 10,000 mAh batteries and data loggers 2, 3, 4, and 5 used 5,000 mAh batteries.



Fig. 4. Soil moisture sensors (left), solenoid valves and pressure sensors (middle), and datalogger (right) in the field.

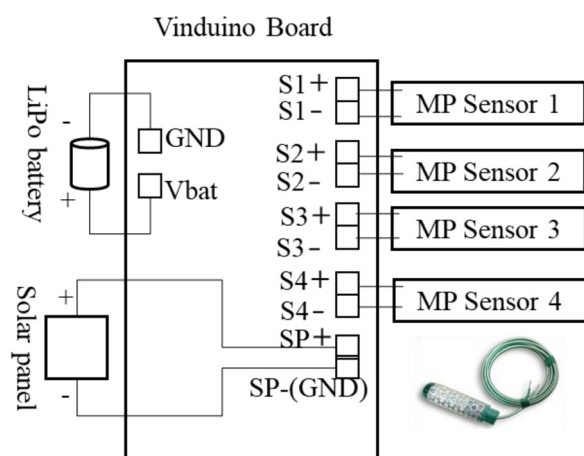


Fig. 5. Connection of the matric potential sensors with the data logger. A data logger has 4 sets of sensor ports (S1–S4) and one power ports SP. Each one has two pins, “+” and “-”. The two wires of a sensor can be connected to either pin since MP sensor has no polarity.

2.3.1. Soil MP sensor system

Fig. 5 shows the connection of the soil MP sensors to the data loggers. In each treatment, four soil MP sensors (Watermark 200SS-5, Irrometer Company, Inc., Riverside, CA) were used, and these MP sensors were connected to one data logger. These four sensors were installed in two experimental sections and at two depths (20 and 40 cm). During the installation, a soil sampler was used to dig a cylinder hole at 40 cm depth. The MP sensors were placed into the hole. The original soil was backfilled, and water was added to ensure close contact between sensors and soil. The average soil MP of two experimental sections at 20 cm was used for starting the irrigation in the treatments MP60 and MP40. The sensors at 40 cm were used as a reference to check if the water went down to the depth after each irrigation or fertigation event. The sensors installed in treatments ET and GesCoN were used only for monitoring the soil moisture levels.

2.3.2. Water pressure sensor system

In the pipeline for each treatment, a pressure sensor was installed behind the solenoid valve to indicate the valve status (on or off). The sensors were calibrated with the pressure gauges installed on the same pipeline. By adjusting the opening of the ball valve, eleven pairs of pressure sensor readings and pressure gauge readings from 0 to 207 kPa were

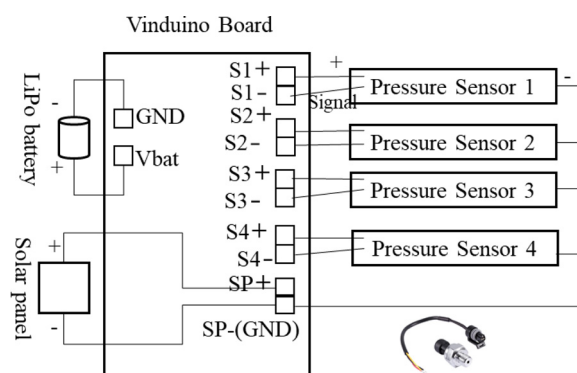


Fig. 6. Connection of the pressure sensors with the data logger. The power input VCC of pressure sensors was connected to the corresponding “+” pins of S1–S4. The pressure sensors output voltage by signal wire was connected to the “-” pins. All the sensor GNDs were connected together to the SP-.

used to calibrate the pressure sensor. Fig. 6 shows the connection of the pressure sensors to the data loggers.

2.3.3. Irrigation valve control

DC latching solenoid valves were used to control irrigation by connecting and disconnecting the water supply. Fig. 7 shows the connection of valves and the data logger. The Vinduino board originally supported only one valve. To control four valves, the sensor ports on the board were used to extend the capability. Sensor ports can send high/low signals at 3.3/0 V, but the current was not enough for executing the valves. Four 2-way relays were added to connect the valves to the main battery. The signals sent to the IN1 and IN2 on each relay could determine the status of the collected valve. When IN1/IN2 is high/low, the valve opens. When IN1/IN2 is low/high, the valve closes. When IN1/IN2 remains the same or no input, the valve does not respond.

2.4. IoT system and data collection

An Internet of things (IoT) system was established to connect the sensors, valves, and data loggers in the field as shown in Fig. 8. A LoRaWAN gateway (Sentrius™ RG191, Laird Technologies, Inc., Chesterfield, MO) was placed in an office at 150 m away from the field. These data loggers were set up as end nodes in the IoT system, which were pre-configured with basic algorithms for sensor data recording and communication. Some modifications were made in the program to fit for our

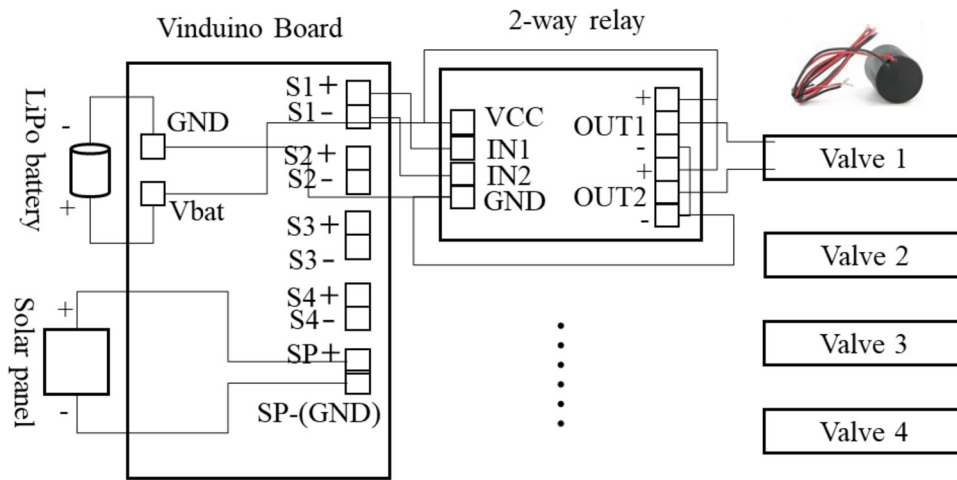


Fig. 7. Connection of solenoid valves with the data logger. One set sensor ports (e.g. S1) send high/low signals to the IN1 and IN2 ports of the 2-way relay to change electric potential at OUT1 and OUT2 to turn on or off the valve.

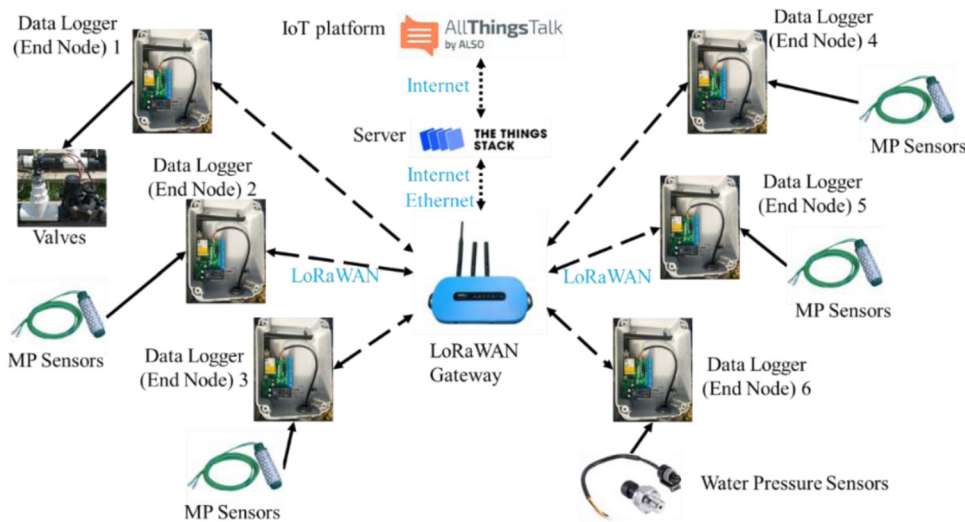


Fig. 8. Structure of the experimental IoT system for the developed irrigation system.

Table 1
Major specifications of the gateway and end nodes used in the system.

Parameters	Gateway	End node (data logger)
Chipset	Semtech Sx1301/1257	GlobalSat LoRa Module LM-513H
Operating band	902–928 MHz	902–928 MHz
Bandwidth	125 kHz	125 kHz
Spreading factor	7	7
Transmitted packet size	6 nodes @10 bytes	10 bytes
Antenna type/gain	Dipole/2 dBi	Whip, Right Angle/2.2 dBi
Transmission power	+27 dBm	+20 dBm

application for sensor data recording and valve control. The gateway and end nodes were configured in a free IoT server ‘The Things Network’, newly named ‘The Things Stack’ (TTS, The Things Network, Amsterdam, Netherlands). The main specifications of the gateway and the end nodes are listed in Table 1. An IoT platform named AllThingsTalk (AllThingsTalk NV, Mechelen, Belgium) was used to store and display the sensor data and control the valves.

To establish the IoT system, the first step was to configure the gateway and the end nodes. The LoRaWAN gateway was connected to a computer with Wi-Fi for matching the parameters, including using TTS as server and ‘915MHz’ as band frequency. In the TTS interface, the gateway, application (the whole irrigation system), and devices (data loggers) was created. The Application Extended Unique Identifier (EUI) and App Key provided by the server were set in the data loggers, establishing the connection of the data loggers and the gateway. Once the

end nodes were powered, they were connected to the TTS after a few seconds.

The next step was to transmit the data to the AllThingsTalk IoT platform and process the data. In TTS interface, AllThingsTalk was integrated to the created application. Then AllThingsTalk was linked to TTS. Sensor data were uploaded and stored in AllThingsTalk for monitoring and irrigation control. Six ‘Devices’ and several ‘Assets’ under ‘Devices’ were added in AllThingsTalk interface corresponding to the six data loggers and the sensors or valves in each data logger. The data communication frequency was applied with a pre-set time interval, which was 1 min. The time for each data transmission duration was 100 ms, which followed the LoRaWAN duty cycle regulation of less than 1% of duty cycle. The sensor data (uplink) and valve control (downlink) were communicated as binary payload. Each value, such as the battery voltage, sensor reading, and valve status was parsed from a byte between 0 and

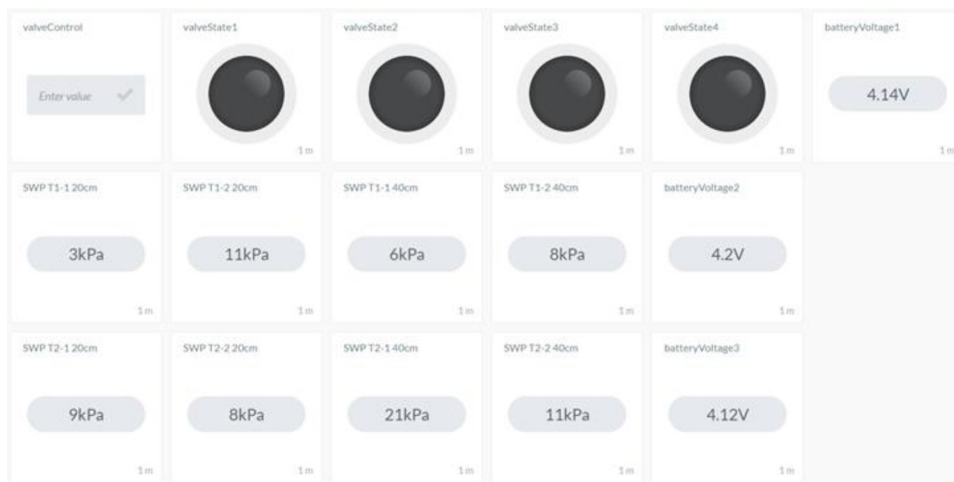


Fig. 9. Sensor data display and irrigation valve control on the IoT platform (part of the whole interface).

255. These were converted to actual values by editing the payload format in the 'Devices' interface. The valves can be separately controlled to turn on or off through AllThingsTalk.

The real-time status of the sensors and valves was displayed on a pinboard (Fig. 9), which is a customized dashboard in AllThingsTalk. The historic data can be displayed, stored, and exported in a comma-separated values (CSV) file. The status of the solenoid valves was shown by a circular indicator with black color for close and green color for open. The valves were controlled to be opened individually or simultaneously. For the two MP-based irrigation treatments, the MP sensor readings were compared with the pre-set thresholds, and a notification can be sent by email or mobile application when the threshold was reached. However, full automation was tested but not applied routinely for the irrigation, instead of a manual control through the IoT platform was used to start and end irrigations. The major reason was that there was signal loss occasionally on the farm which could lead to a delayed response of valves with automatic setting.

2.5. IoT irrigation system performance evaluation

The IoT precision irrigation system was evaluated throughout the growing season. The IoT system function was evaluated, including connectivity and battery usage. The robustness of valve control was also tested. Soil moisture from all the sensors was recorded and closely monitored. The water usage was recorded after the last harvest by reading the flow meter. The crop was harvested five times throughout the growing season from August 7, 2020, to September 22, 2020. Harvests were conducted on ten representative plants selected in the central row of each experimental unit, tomatoes were harvested as single fruits as they turned color, following standard harvesting practices [29], and marketable fruit yield was recorded for each treatment. Irrigation water use efficiency (iWUE) was calculated by dividing the marketable fruit fresh weight by the total water usage for each treatment. Analysis of variance (ANOVA) was applied to marketable fruit yield and iWUE for the four treatments. Statistical analysis was performed using the MIXED and GLIMMIX procedures in SAS 9.4 (SAS Institute Inc., Cary, NC, USA., 2016). All means were compared by the least significant difference (LSD) test at $P \leq 0.05$.

3. Results and discussion

3.1. Feasibility of the IoT system

In general, the IoT system worked properly during the entire irrigation period. There were only a few occasions of data loss in the IoT system. Six data loggers had an average data loss rate of 5.51%. The

gateway was placed inside a building without a window directly facing to the field, which may have led to some signal loss. This can be improved if the gateway is installed outside with less occlusion. The disconnection of any component such as the gateway, the server The Things Network, and the IoT platform AllThingsTalk contributed to signal loss. However, the sensing system was not affected significantly since data were uploaded at the high frequency of every minute without long period of data loss. During the experiment, a few instances of data losses lasted more than 10 min were observed. If the irrigation was applied during these periods, it is possible that the valves would not respond on time.

The batteries in the six data loggers were charged by solar panels and maintained sufficient power throughout the three-month experiment period. Fig. 8 shows the voltage of these batteries over time. There were a few occasions of voltage drop for all data loggers due to continuing cloudy and rainy weather. Solar panel wires were inadvertently disconnected for data logger 3 between 60 and 70 days after transplanting (DAT), resulting in an obvious voltage drop. Overall, the battery voltages were above 4.0 V throughout the experiment, which showed that the 5,000 mAh LiPo battery with a solar panel could support a data logger working continuously in the open field (Fig. 10).

3.2. Valve control

During the experiment, irrigation was applied successfully by controlling the status of the solenoid valves with the implemented IoT system. When the soil moisture threshold was reached, the platform sent a notification through the mobile application and an email to the end-user. The operator then turned the corresponding valves on, and the irrigation started. When the valves were opened, the status of the valves showed 'Green' on the IoT interface, and the pressure of the water line indicated a positive reading. When the valves were closed after the given time of each irrigation event, the valve status showed 'Black', and the reading of the pressure sensors returned to zero. It was also observed that the valve status changed to 'Green' while the pressure sensor reading did not change accordingly. In this situation, the valve did not actually open with pressurized flowing water. In other words, the solenoid did not respond even if the signal was received by the controller. By sending the signal several times, the valve did finally open or close. This can be attributed to a clogging in the solenoid or a small voltage used for powering the valves (4.0 V). In the future, a higher voltage (9.0 V) will be used to improve the robustness of the valve. A remote control with manual operation was used in the experiment, while for a fully automatic irrigation system, the start and end of an irrigation cycle need to be programmed into the controller according to the soil moisture thresholds.

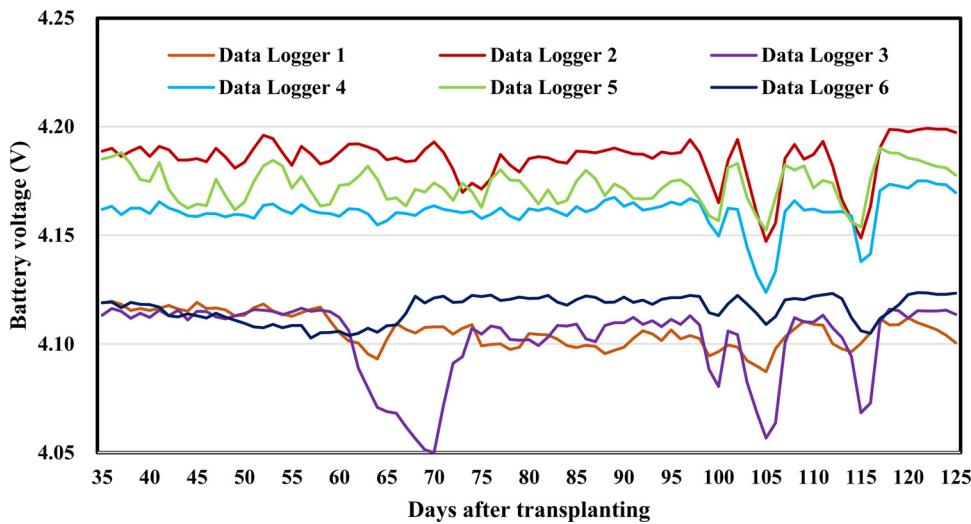


Fig. 10. The battery voltage of the six data loggers during the experiment.

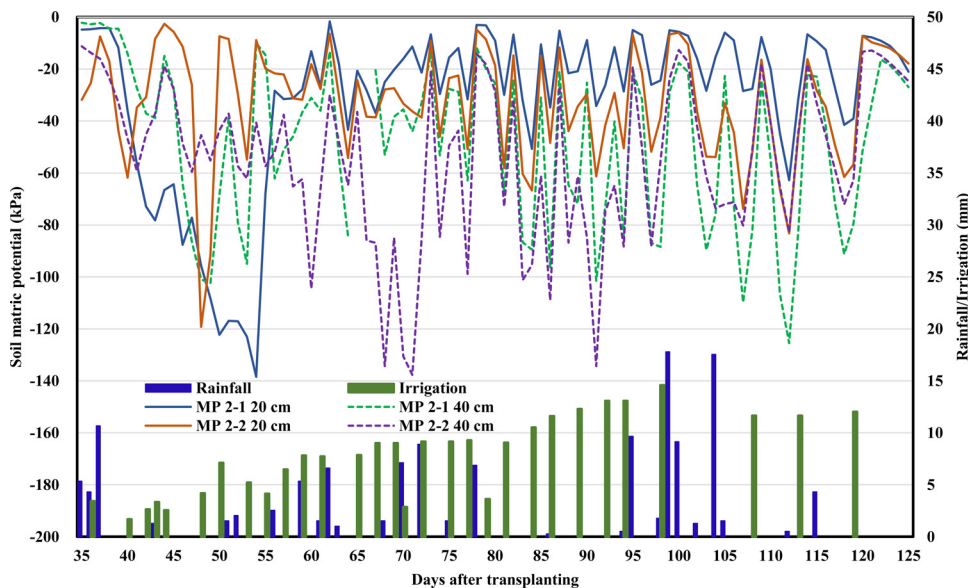


Fig. 11. Daily average soil MP sensor readings, rainfall, and irrigation of treatment MP60 during the experiment.

3.3. Soil moisture monitoring with the IoT system

Generally, the sensor readings represented the changes in the soil moisture level in the test field. When irrigation or relevant rainfall events occurred, the sensor readings responded accordingly, showing increased soil moisture levels. For example, Fig. 11 illustrates the daily average soil MP60 sensor readings, rainfall, and irrigation of four treatments between 35 and 125 DAT. When the average sensor reading at 20 cm reached -60 kPa, irrigation started. Thus, the daily average readings of sensors at 20 cm were usually above -60 kPa during the experiment. The sensor readings at 40 cm indicated that water reached that depth during the irrigation events, but the soil was not too wet, which proved that the irrigation was sufficient but not in excess. For this treatment, we observed one sensor failure at the beginning of the experiment, which is shown by the MP reading that reached large negative values during 45–55 DAT in one of the 20 cm locations. After replacing the malfunctioning sensor, the normal soil matric potential data resumed.

Fig. 12 shows the sensor readings around an irrigation event of treatment MP60 at 77 DAT. The irrigation was applied based on the average readings of two sensors at 20 cm depth. When the average readings reached -60 kPa at 18:50, the notification was sent, and the valve was opened remotely. The increased water pressure readings showed that

the water was on. The MP sensor readings increased after 30 min and stopped increasing after about 2 h. Sensors at 40 cm and at the first installation position responded first. Commonly, water reached 20 cm first, then 40 cm. But the sensors were not close to the drip tapes, and the irrigation water spread in both the vertical and horizontal direction to reach the sensors. For this reason, sometimes the irrigation water reached the 40 cm sensors first. At 22:20, the irrigation ended, and the water pressure readings went back to 0. Before the irrigation, the sensors readings at 20 cm were near -60 kPa, and the sensor readings at 40 cm were at -100 and -140 kPa. During the irrigation, all the sensor readings gradually reached the soil matric potential values above -20 kPa, while sensors at 40 cm were still dryer than sensors at 20 cm. There was a slight variation for the MP 2-1 40 cm sensor after irrigation, which may be due to the accuracy of the sensor itself. Sensors at 40 cm depth usually would become dryer faster than sensors placed at 20 cm depth.

3.3. Evaluation of the irrigation water use efficiency and crop yield performance

Table 2 shows the total water usage per unit area, the fruit marketable yield, and iWUE of the four treatments. Treatment MP40 used 30.6% less irrigation water compared to the ET treatment, while there

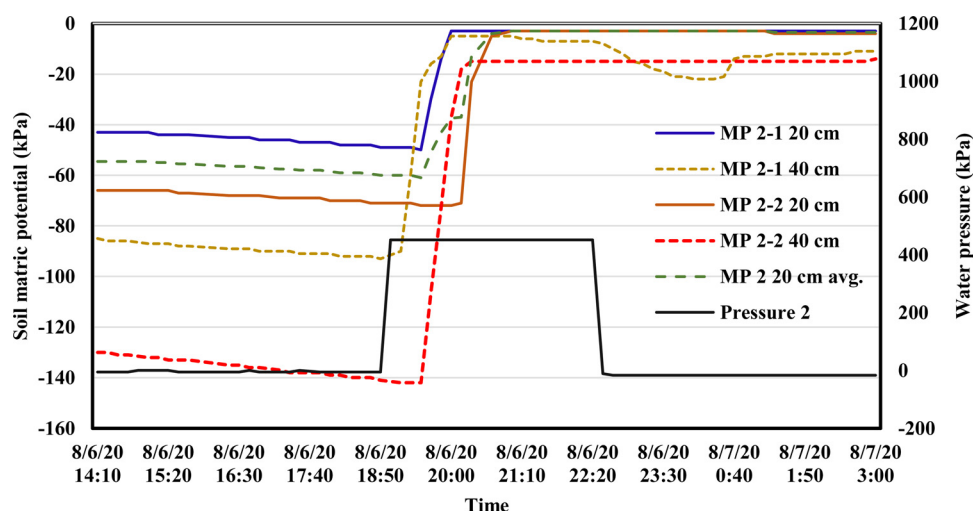


Fig. 12. Soil MP sensors and pressure sensor readings of treatment MP60 during an irrigation event.

Table 2

Total irrigation water usage per area and irrigation water use efficiency (iWUE)¹.

Treatments	Water Usage (m ³ ha ⁻¹)	Marketable Yield (Mg·ha ⁻¹)	iWUE (kg·m ⁻³)
ET	2440	54.21 bc	22.22 b
MP60	2357	62.43 ab	26.49 ab
MP40	1695	47.34 c	27.94 a
GesCoN	2339	66.19 a	28.38 a
P-value	Na	0.01	0.05

¹ Means followed by different letters within the column are significantly different at $P \leq 0.05$ by the least significant difference (LSD) t-test.

was no significant difference in marketable fruit yield between MP40 and ET. However, MP40 marketable yield was significantly lower compared to that of MP60 and GesCoN. ET recorded the highest irrigation water usage, and MP60 and GesCoN followed by using 3.4 and 4.1% less irrigation volume compared to ET. At the same time, GesCoN and MP60 provided 22.1 and 15.2% higher marketable yield compared to ET plots. Tomato fruit marketable yield was highest in plots managed with GesCoN, and lowest in plots managed with MP40. MP60 provided marketable yield comparable to both GesCoN and ET, and higher compared to MP40. ET managed Plots provided marketable yield lower than GesCoN, while not statistically different from MP40. Plants irrigated based on MP40 and GesCoN had 25.7% and 27.7%, respectively, higher iWUE compared to those irrigated based on ET. The iWUE of MP60 plots was not different from those of all the other treatments. However, considering the lower marketable yield observed in MP40 plots, its high iWUE was not advantageous compared to MP60 and GesCoN.

Overall, the results revealed that the IoT irrigation system performed well and could be an excellent tool for the precision irrigation management of plasticulture fresh-market tomato. The batteries for the data loggers were charged sufficiently by the attached solar panels. A series of data were recorded and displayed for the soil matric potential sensors and pressure sensors. Irrigation was applied successfully by controlling the solenoid valves based on the soil moisture level. Even though no significant negative effects were observed in the sensor data monitoring and valve control, the IoT system had a data loss rate of 5.5% on average, which may be attributed to the indoor gateway placement and disconnection of each component in the network. In future studies, more attention will be given to the signal strength and path loss to assist addressing the signal loss issues. High frequency of data communication (every minute) largely mitigated this problem. Meanwhile, it was observed that the valves did not respond with the first sent signal on some occasions, which was possibly caused by the low voltage power used for the valves. A higher voltage power supply is suggested for future system design. The marketable fruit yield and iWUE were analyzed and com-

pared for all four treatments. MP40 used much lower irrigation volume than other treatments likely because of improper installation and positioning of the soil matric potential sensor. The other three treatments used similar irrigation volume, nevertheless, the higher marketable fruit yield obtained with GesCoN compared to ET could be explained by a better distribution of irrigation water throughout the growing season and potentially by lower loss of nutrients. The gain in marketable yield while saving water observed in both GesCoN and MP60, compared to ET-based irrigation management, is consistent with the results of previous studies. Therefore, we can suggest both methods have the potential to improve vegetable crop water use efficiency and sustainability. Moreover, the IoT-based system developed using LoRaWAN technology and employing MP sensors can be potentially used for precision and automatic irrigation applications in vegetable production systems, and could be integrated with the GesCoN that currently does not allow automatic irrigation and/or fertigation.

To effectively use sensor-based irrigation, an adequate number of sensors should be installed in representative areas across the field. If the sensors installed do not represent the average condition of the field, as was observed with MP40, using multiple sensors can mitigate the error induced by malfunctioning or misplaced soil moisture sensors. It is important for the sensors to provide representative readings to ensure the effectiveness of the sensor-based irrigation. DSS systems such as GesCoN, on the other hand, require the use of a nearby weather station and the development and calibration of the water balance model for specific crops and environments. If the models at the base of the DSS are not accurate, the DSS-based irrigation cannot assure optimal crop performances. While the GesCoN does not require the use of soil moisture sensors, in the future the two methods could be combined. The sensors' readings can be used to double-check the recommendations provided by the DSS. Moreover, the sensors' readings can be used to validate and improve the accuracy of the DSS model.

In this experiment, the irrigation was controlled manually through the IoT platform. The signal loss of the system and no response of valves

limited the possibility to apply automatic irrigation control. Future improvements will be made with adding automatic irrigation rules to the IoT platform or in the data logger to achieve fully automatic irrigation.

4. Conclusion

An IoT-based precision irrigation system with LoRaWAN technology was developed and evaluated in a plasticulture fresh-market tomato field. The following major conclusions can be drawn from the study:

- (1) In general, the system worked effectively for irrigation application in the tomato field. The soil moisture levels were monitored and recorded throughout the growing season. The irrigation was applied successfully by controlling the solenoid valves based on the soil moisture level. The batteries for the data loggers were charged sufficiently by the attached solar panels.
- (2) The IoT system provided real-time soil moisture data and indicated that the scheduled irrigations for the matric potential treatments are sufficient.
- (3) Among tested treatments, MP40 provided the worst performance in terms of marketable yield, while GesCoN provided the best performance increasing yield (22.1%) and iWUE (27.7%) compared to ET-based irrigation scheduling. MP60 provided the second-best performance, although marketable yield and iWUE were not significantly different from both GesCoN and ET-based irrigation scheduling.

The results suggest that the developed IoT-based system using LoRaWAN technology can be potentially used for precision and automatic irrigation applications and improve the sustainability of vegetable production systems enhancing marketable yield while saving irrigation water. In the future, using outdoor gateway and higher voltage batteries for driving the solenoid valve may further improve the LoRaWAN IoT based irrigation system.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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